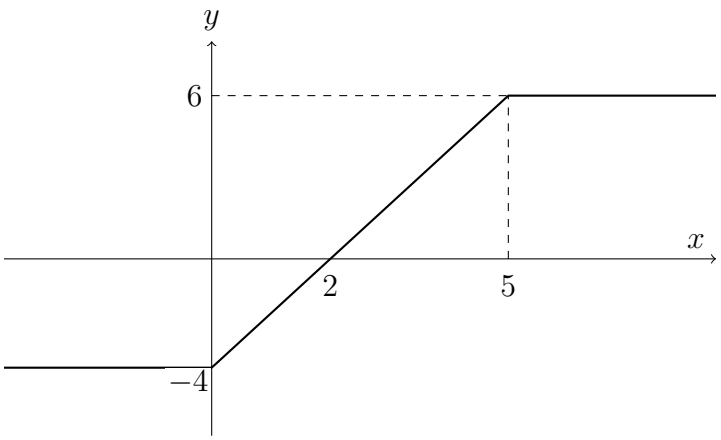
(i)

$$x \le 0: \quad -x-(5-x)+1=-4$$

$$0 \le x \le 5: \quad x-(5-x)+1=2x-4$$

$$x \ge 5: \quad x-(x-5)+1=6$$

$$\therefore \boxed{\mathrm{f}(x)=\begin{cases}-4 & x \le 0\\ 2x-4 & 0 \le x \le 5\\ 6 & 5 \le x\end{cases}}$$



(ii) Let $\mathrm{g}(x)=\alpha x+\beta|x|+\gamma|x-5|+\delta$
The gradients on the three pieces are

$$x<0: \ \alpha-\beta-\gamma=-1, \quad 0<x<5: \ \alpha+\beta-\gamma=3, \quad x>5: \ \alpha+\beta+\gamma=1$$

$$\implies \beta=2, \quad \gamma=-1, \quad \alpha=0$$

$$\mathrm{g}(0)=0: \ 5\gamma+\delta=0 \implies \delta=5$$

$$\therefore \boxed{\mathrm{g}(x)=2|x|-|x-5|+5}$$

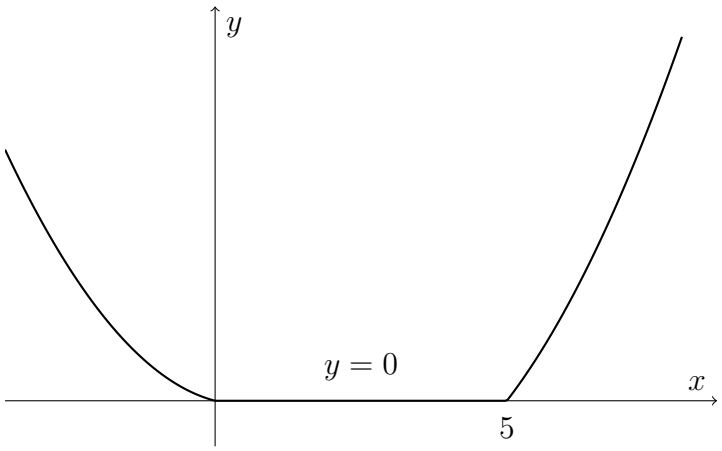
(iii) When $x \le 0$ or $x \ge 5$, $x(x-5) \ge 0$

$$x \le 0: \quad x^2-x+4x+x^2-5x=2x^2-2x$$

$$0 \le x \le 5: \quad x^2 - x - 4x + 5x - x^2 = 0$$

$$x \ge 5: \quad x^2 - x - 4x + x^2 - 5x = 2x^2 - 10x$$

$$\therefore \boxed{\mathrm{h}(x) = \begin{cases} 2x(x-1) & x \le 0 \\ 0 & 0 \le x \le 5 \\ 2x(x-5) & 5 \le x \end{cases}}$$



(iv) As in part (iii), $|x(x-5)| = x(x-5)$ for $x \le 0$ and for $x \ge 5$, and $|x(x-5)| = -x(x-5)$ for $0 \le x \le 5$. Consider

$$\mathrm{k}(x) - x^2 + |x(x-5)|$$

$$x \le 0: \quad 10x - x^2 + x(x-5) = 5x$$

$$0 \le x \le 5: \quad 2x^2 - x^2 - x(x-5) = 5x$$

$$x \ge 5: \quad 50 - x^2 + x(x-5) = 50 - 5x$$

This is piecewise linear with gradient 5 for $x < 5$ and -5 for $x > 5$, so as in part (ii) it has the form $a + b|x-5|$

$$x < 5: \ -b = 5; \quad x > 5: \ b = -5$$

$$x = 5: \ a = 25$$

$$\therefore \boxed{\mathrm{k}(x) = x^2 - |x(x-5)| + 25 - 5|x-5|}$$

$x \leq 0$: $|x(x-5)| = x^2 - 5x$ and $|x-5| = 5 - x$

$$x^2 - \big(x^2 - 5x\big) + 25 - 5(5-x) = 10x \checkmark$$

$0 \leq x \leq 5$: $|x(x-5)| = 5x - x^2$ and $|x-5| = 5 - x$

$$x^2 - \big(5x - x^2\big) + 25 - 5(5-x) = 2x^2 \checkmark$$

$x \geq 5$: $|x(x-5)| = x^2 - 5x$ and $|x-5| = x - 5$

$$x^2 - \big(x^2 - 5x\big) + 25 - 5(x-5) = 50 \checkmark$$

This solution was submitted by **Jon**
Published and refined by **Jon**

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